SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.2: The Cell — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 1.2: The Cell
Driving Question	DRIVING QUESTION: What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Predict & Anchor Phenomenon — Microscopes as Windows into the Cell**	Students examined a fresh sukuma wiki leaf under a light microscope at increasing magnifications (4×, 10×, 40×). They observed that the leaf, which appears as a uniform green sheet to the naked eye, is composed of discrete cells, each bounded by a wall and containing visible internal structures (chloroplasts, a large vacuole, and a nucleus). Pedagogical note: Cold-call at least 4 students to describe one thing they see at each magnification before naming structures — this surfaces prior knowledge and misconceptions about what 'inside a cell' means.	Students learned that (1) all living things, including familiar Kenyan food crops like sukuma wiki, are made of cells; (2) cells contain distinct internal structures not visible to the naked eye; (3) the light microscope is the tool that makes this level of organisation visible; and (4) increasing magnification reveals progressively more internal detail. The anchor phenomenon — one sukuma wiki leaf cell keeping the whole plant alive — was introduced and posted on the Driving Question Board (DQB).	Students began to explain that the visible green structures (chloroplasts) must be doing something important for the plant, and that the boundary (cell wall/membrane) separates the cell's interior from its surroundings. Teacher should note that explanations at this stage are intentionally incomplete — the goal is to generate productive questions, not closure. Redirect any student who jumps to memorised definitions back to the evidence: 'What in your slide makes you say that?'
Lesson 2 **Light vs. Electron Microscopes — Revealing Hidden Structures in the Sukuma Wiki Cell**	Students compared side-by-side images: a light micrograph and a transmission electron micrograph (TEM) of a plant cell similar to sukuma wiki. They recorded observable differences in the number of structures visible, sharpness of membranes, and internal detail of organelles (e.g., the double membrane of the chloroplast and mitochondria are invisible under the light microscope but clear under the TEM). Pedagogical note:	Students learned that (1) the light microscope uses visible light (wavelength ~400–700 nm) and has a resolution limit of ~200 nm; (2) the electron microscope uses a beam of electrons (wavelength ~0.004 nm) achieving resolution of ~0.1–0.2 nm; (3) this difference in resolution is why organelles such as ribosomes, smooth ER, and the nuclear envelope's double membrane are only visible under the electron microscope; and (4) electron	Students explained why Kenyan researchers at institutions such as the Kenya Medical Research Institute (KEMRI) use electron microscopes when studying cell-level disease mechanisms — the structures relevant to disease (e.g., viral particles, membrane damage) are below the resolution of the light microscope. Teacher should connect this back to the driving question: 'If we can only see some organelles with an electron microscope,

	Provide a structured observation table with columns 'Seen under LM,' 'Seen under EM only,' and 'Still unclear' — this scaffolds disciplinary literacy and surfaces gaps for the DQB.	microscopes require specimens to be dead and in a vacuum, so living processes must be inferred from fixed images.	how did scientists know the sukuma wiki cell had those structures at all?' Allow 30 seconds of wait time before cold-calling.
Lesson 3 **Animal Cell Ultrastructure: Observing the Hidden Machinery of Life**	Students observed annotated TEM images of a human cheek cell (a relatable Kenyan context — 'the same cells lining your own mouth'). They identified and labelled at least 9 organelles: nucleus (with nuclear envelope and nucleolus), mitochondria, rough ER, smooth ER, Golgi apparatus, ribosomes, lysosomes, cell membrane, and cytoplasm/cytosol. Pedagogical note: Use a progressive-reveal strategy — display the TEM image unannotated first and have students circle structures they can distinguish before revealing labels. Cold-call at least 5 students to justify their identification from structural evidence (shape, location, membrane layers).	Students learned the structure–function relationship for each of the 9 organelles in an animal cell: (1) nucleus — stores DNA and directs cell activity; (2) mitochondria — site of aerobic respiration producing ATP; (3) rough ER — synthesises and transports proteins (studded with ribosomes); (4) smooth ER — synthesises lipids and detoxifies; (5) Golgi apparatus — packages and dispatches proteins and lipids; (6) ribosomes — site of protein synthesis; (7) lysosomes — contain digestive enzymes for waste breakdown; (8) cell membrane — selectively permeable boundary; (9) cytoplasm — medium suspending organelles and site of many reactions.	Students explained how organelles work as an integrated system using a Kenyan analogy: the cell is like a busy market in Gikomba — the nucleus is the market manager issuing instructions, ribosomes are the producers, the ER is the transport corridor, the Golgi is the packaging and dispatch unit, and lysosomes are the waste collectors. Teacher should prompt: 'Which of these organelles would be especially busy in a muscle cell of a Kenyan marathon runner like Eliud Kipchoge, and why?' (Expected: mitochondria — high ATP demand.) Allow 45 seconds wait time.
Lesson 4 **Plant Cell Ultrastructure: What Extra Structures Does the Sukuma Wiki Cell Have?**	Students examined annotated TEM images of a sukuma wiki (kale) leaf mesophyll cell and compared them directly to the animal cell images from Lesson 3 using a Venn diagram. They identified three structures present in the plant cell but absent in the animal cell: cell wall (cellulose), chloroplasts (with visible thylakoid membranes and grana), and a large central vacuole. They also noted that plant cells lack lysosomes (the vacuole performs similar functions) and centrioles. Pedagogical note: The Venn diagram comparison is a high-leverage activity — circulate and check that students are using structural evidence from the images, not recalled text, to populate the diagram.	Students learned that (1) the cellulose cell wall provides mechanical support and shape to plant cells, explaining why sukuma wiki leaves are rigid; (2) chloroplasts are the site of photosynthesis — they contain chlorophyll in thylakoid membranes organised into grana, and a fluid stroma where the Calvin cycle occurs; (3) the large central vacuole stores water, pigments, and waste products, and its turgor pressure is what keeps the sukuma wiki plant upright; and (4) the presence of a middle lamella and plasmodesmata connects adjacent plant cells, enabling communication across the whole leaf.	Students explained the driving question more fully: a sukuma wiki plant stays alive because its cells have chloroplasts (capturing sunlight energy), a cell wall (maintaining structure), and a vacuole (regulating water balance) — structures an animal cell lacks. Teacher should ask: 'If you leave sukuma wiki in the sun without water, it wilts. Which organelle or structure is most directly responsible, and what is happening at the cellular level?' This connects cell biology to everyday Kenyan agricultural observation. Record updated explanations on the DQB.
Lesson 5 **The Nucleus and Genetic Control: The Cell's Command Centre**	Students observed high-magnification TEM images of the nucleus, focusing on (1) the double nuclear envelope with visible nuclear pores, (2) the nucleolus (dense region for rRNA production), and (3) chromatin (diffuse) versus condensed chromosomes in a dividing cell. They also	Students learned that (1) the nucleus is bounded by a double membrane (nuclear envelope) perforated by nuclear pores that regulate molecular traffic; (2) DNA is organised with histone proteins into chromatin, which condenses into chromosomes during cell division; (3) the	Students explained how the nucleus acts as the control centre linking back to the driving question: every function performed by every organelle in a sukuma wiki or human cell is ultimately directed by instructions encoded in nuclear DNA. Teacher should pose: 'A cheek cell and a liver cell in the same

	viewed a short animation showing mRNA exiting through a nuclear pore to a ribosome on the rough ER. Pedagogical note: Many students conflate 'nucleus' with 'nucleolus' — use cold-calling with image-pointing ('Point to the nucleolus. Now point to a nuclear pore.') to check accuracy before moving to function.	nucleolus is the site of ribosomal RNA (rRNA) synthesis — cells with high protein production needs have large, prominent nucleoli; and (4) genetic information flows from DNA → mRNA (transcription in nucleus) → protein (translation at ribosomes in cytoplasm), establishing the central dogma as the mechanism by which the nucleus controls all cell activity.	Nairobi hospital patient contain identical DNA. Why do they look and behave so differently?' (Expected: differential gene expression.) This previews cell specialisation content and should be added as an open question to the DQB.
Lesson 6 **Organelle Teamwork — Protein Secretion Pathway and Cellular Organisation**	Students traced the journey of a secretory protein (using a labelled diagram and a step-sequencing card sort) from ribosome on the rough ER → ER lumen → transport vesicle → Golgi apparatus (cis to trans face) → secretory vesicle → cell membrane → extracellular space. They annotated each step with the organelle responsible and the energy cost (ATP). A Kenyan context was used: 'Imagine this protein is insulin being made by a pancreatic cell in a diabetic patient at Kenyatta National Hospital.' Pedagogical note: The card sort is the core activity — do not rush to whole-class reveal. Give groups 8 minutes, then have each group justify one step to the class before correcting.	Students learned that (1) organelles do not act in isolation — protein secretion requires coordinated action across at least five compartments; (2) vesicles are membrane-bound transport sacs that carry cargo between organelles without mixing compartment contents; (3) the Golgi apparatus modifies, sorts, and addresses proteins to their correct destination (secretion, lysosome, or membrane insertion); and (4) this endomembrane system (ER + Golgi + vesicles + cell membrane) forms a functionally integrated network, illustrating that cell structure and function are inseparable.	Students explained that the complexity of organelle coordination answers why a cell — not just a bag of chemicals — is needed to sustain life. Returning to the driving question: even a single sukuma wiki leaf cell must coordinate dozens of simultaneous processes (photosynthesis, protein synthesis, waste disposal, water regulation) through organelle teamwork. Teacher should ask students to update their cell models on the DQB to show arrows representing communication and transport between organelles, not just isolated structures.
Lesson 7 **3D Cell Model Project — Building and Justifying a Model of a Plant or Animal Cell**	Student groups constructed physical 3D models of either a plant cell (sukuma wiki) or an animal cell (human cheek cell) using locally available Kenyan materials — maize kernels, beans, groundnuts, bottle caps, foil, clay, fabric — selected deliberately to represent specific organelles based on structural analogies. Each group presented their model, pointing to each component and explaining (a) which organelle it represents, (b) what structural feature of the material mirrors the organelle's actual structure, and (c) what function that organelle performs in the living cell. Pedagogical note: Assess using a rubric with three columns: accuracy of structure, accuracy of function, quality of justification. Do not accept 'the nucleus is the bean because it is the control centre' without a structural reason (e.g., 'the bean has a hard	Students demonstrated consolidated knowledge of (1) the names and relative spatial arrangements of major organelles in both plant and animal cells; (2) the structural basis for each organelle's function (e.g., the folded inner membrane of mitochondria = large surface area for ATP production); (3) the distinguishing features between plant and animal cells (cell wall, chloroplast, large vacuole vs. centrioles, lysosomes); and (4) how to use a physical model as a scientific communication tool, including acknowledging the model's limitations (e.g., 'our model cannot show the dynamic movement of vesicles').	Students gave their most complete explanation yet of the driving question: a sukuma wiki plant or a human being in Nairobi stays alive because each cell contains a coordinated set of membrane-bound organelles that divide metabolic labour — capturing energy, synthesising proteins, managing waste, and maintaining boundaries — all under genetic direction from the nucleus. Teacher should highlight model limitations as a scientific practice: 'What does your 3D model fail to show that a real cell does?' This is a formative assessment checkpoint before Lesson 8.

	outer coat like the nuclear envelope').		
Lesson 8 **Cells, Tissues, Organs, and Organisms — From the Sukuma Wiki Cell to the Whole Living Thing**	Students examined a sequence of images at increasing scales: (1) TEM of a single sukuma wiki mesophyll cell, (2) light micrograph of mesophyll tissue (many cells), (3) cross-section of a sukuma wiki leaf (showing multiple tissue layers: epidermis, palisade mesophyll, spongy mesophyll, vascular bundle), (4) a photograph of a whole sukuma wiki plant, and (5) a photograph of a Kenyan farmer's field in Kiambu County. They mapped each level onto the hierarchy: cell → tissue → organ → organism → population. Pedagogical note: Ask students to identify which level they have studied in depth (cell) and which levels were implied but not examined — this grounds the lesson in metacognitive awareness of the sub-strand's scope.	Students learned that (1) cells are the fundamental unit of life but do not act alone — similar cells are organised into tissues (e.g., palisade mesophyll tissue in sukuma wiki); (2) tissues are organised into organs (the leaf), organs into organ systems (shoot system), and organ systems into the organism; (3) each level of organisation has emergent properties not present at lower levels (a single chloroplast cannot perform the gas exchange that a whole leaf does); and (4) the same hierarchical principle applies to the human body — a single cheek cell to the tissues, organs, and organ systems of a person living in Nairobi.	Students delivered their final, evidence-based answer to the driving question: what makes up an organism is a hierarchy of organisation beginning with the cell. The tiny structures inside a single cell — each with a specific structure enabling a specific function — collectively sustain cellular life; cells organised into tissues and organs sustain organ-level function; and organ systems working together sustain the whole organism, whether a sukuma wiki plant or a human being. Teacher should facilitate a final DQB review: 'Which questions we posted at the start can we now fully answer? Which remain open?' Unanswered questions should be acknowledged as the frontier — the basis of ongoing biological research.